‘Scam PACs’: A Visual Analysis of Campaign Finance Fraud

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1 Introduction

Election outcomes in representative government frequently correlate with monetary investments into campaigns [2], motivating citizens to donate personal resources to a candidacy. In the United States, this is done through the work of Political Action Committees (PACs)— organizations that spend money donated by individuals, businesses, and other groups to influence elections. This system is unfortunately easy to exploit, resulting in the existence of scam PACs (SPACs) which solicit donations from the public under the guise of raising money in support of a candidate. However, the funds never reach the campaign for which it was raised. Instead, these donations are used in the self-interest of the SPAC founders.

A cursory look through the history of campaign finance provides an understanding of how the system allows SPACs to flourish. The holes in the system began to grow with the landmark case, *Buckley v. Valeo*. Following the Watergate scandal in the early 1970s, Congress— recognizing the need to regulate corruption in campaign finance—passed the Federal Election Campaign Act (FECA)[9]. The Act and its subsequent amendments placed limits on campaign fund contributions and expenditures and implemented disclosure requirements. This was quickly contested in *Buckley*, which quickly rendered the amendments only partially effective when the Supreme Court upheld contribution limits while striking down limits on expenditures. Contribution limits were deemed constitutional as they seemingly promote campaign fairness without infringing on citizen’s rights to participate in the political process [9]. The same logic was not applied to expenditure limitations, which were said to be in violation of the first amendment[7].

Unsurprisingly, campaign finance corruption did not disappear after this series of amendments— it only shifted. Nearly two decades after *Buckley*, the notable Bipartisan Campaign Finance Reform Act was passed, which placed constrictions on contribution sources in federal elections [7][4]. Congress and the Supreme Court have revisited aspects of campaign finance regulations many times over the years, but the campaign expenditure landscape has not shifted much from the FECA amendments of the 1970s. Today, provided a political committee is properly registered and discloses where money was spent, it is not required to reveal further recipient or vendor details, including the vendor owner or actual work performed by the vendor [16][20]. In short, the flexibility allows SPACs to thrive. They can register with the Federal Election Commission (FEC), disclose their vendors with no further accountability, and use donations as they like.

SPACs interfere with the trust citizens place in campaign organizations, rob citizens of the future in which they hope, and diminish confidence in a representative government [9]. Burns et al. estimate that annually over 5% of "cognitively intact, community-dwelling older adults" are victims of financial scams[5]. While SPACs are not directly mentioned in the article, the statistic highlights a portion of the vulnerable population we seek to protect by creating a resource in an accessible format to convey data made publicly available by the FEC.

2 Problem Definition

The FEC collects and discloses campaign finance data to the public openly. This data, however, is vast and complex. Organizations such as OpenSecrets attempt to make the data more digestible by providing articles, charts, tables, and graphs. To date, however, no tool exists for the average citizen to be able to easily visualize connections between PACs and their vendors, much less indicate whether a PAC could be an undisclosed SPAC. We seek to make this data accessible to the public by creating an interactive web interface for users to explore PAC spending. The user can search for a specific PAC and view focused data about the PAC, gaining insight into its character.

Innovation goals for this study are:

- Displaying PAC expenditure data as an interactive, easily-searchable network between PACs and vendors. Current visualizations are limited in scope and focused on contributions.

- Attempting an experimental labeling approach by using an unsupervised ML model. SPAC labeling is currently done manually [13].
- Providing a novel SPAC detection indicator by scoring a PAC on scam likelihood. This does not exist.
- Using metrics derived from the network graph as inputs to the ML models in addition to PAC features. Leading studies solely use individual PAC attributes[13].

3 Literature Survey

The literature on strictly detecting SPACs is sparse—netting only one relevant hit when searching for "detecting scam PACs" since 2020, and that an unpublished study [13]. However, an extensive amount of research is available for review under the general scope of both of campaign finance and identifying financial fraud.

3.1 Network Analysis

Financial fraud never occurs in isolation. The nature of money is that it bears an origin and flows towards a destination, creating a web of connections. As most transactions occur as numerical transfers to accounts, the created connections make network analysis a popular tool for researchers seeking to develop ways to find fraud. For example, Singh and Best created a prototype application for flagging potential money-laundering using a node-linked analysis, demonstrating the effectiveness of network visualization to detect dubious financial patterns [17]. Political science studies also prove that network graphs ease analysis in topics such as party spending[11], donations, and historical networks[10] and highlight useful relationships not immediately obvious from looking at individual entities [19]. The graphs built for these research projects, while utilizing the FEC data, were stand-alone analyses designed to answer specific questions. Unfortunately, their results are not easily explored by other researchers or the public.

3.2 Machine Learning

To identify scams, fraud, and bad money, researchers unsurprisingly reach for machine learning algorithms. Commonly used methods include logistic regression[21], support vector machines[12], ensemble-learning[14][3], and artificial neural network models[1]. In 2021, Li Zhao, recognized the growing need for SPAC detection and conducted an unpublished study [13] employing a random forest model for this purpose. However, a key challenge was the reliance on manual SPAC identification. This requires an understanding of PAC characteristics and extensive research into each individual PAC in order to correctly label SPACs in order to train the supervised model.

To keep up with fraud evolution, many researchers are combining algorithms to improve accuracy. For instance, Xu used unsupervised mapping and random forest to improve the robustness of credit risk assessment[22]. Craja, Kim, and Lessmann[8] developed a novel deep learning model combining textual and financial data to detect financial statement fraud. Similarly, Somkunwar et al.[18] combined network graphing with unsupervised machine learning to achieve a 94.8% accuracy in accounting fraud detection. Though detection was accomplished in these studies, they lack a visualization component that is central to our project objectives.

4 Proposed Methodology

4.1 Intuition

We are combining aspects of reviewed studies to create a tool that average citizens can easily interpret. A directed, non-bipartite network graph forms the skeleton of an interactive webpage highlighting cash flow between entities, where nodes are PACs and their vendors, and edges represent expenditures. To avoid visually overwhelming the user, a search function can navigate to a PAC of interest on our webpage to view connections in a focused subgraph. Interacting with the nodes reveals PAC features, including the scam likelihood, top vendors for the PAC and associated costs.

Algorithmically, the goal is two-fold. The first goal is to label SPACs. Historically, this has been a tedious process requiring intensive research across many sources as the information is not readily available [13]. In fact, OpenSecrets identified 86 SPACs from the 2022 federal election cycle but did not disclose the full list [6]. However, as SPACs are theoretically outliers, an unsupervised machine learning algorithm may be able to identify SPACs. A k-means clustering algorithm (k=2) will be trained using PAC features (such as expenditure categories and corresponding amounts) and graph metrics. SPACs are expected to be outliers in the network graph, as they are less likely to have many connections to other PACs or shared vendors, therefore making

graph metrics (e.g. eigenvector centrality) potentially significant features in the model.

Providing the labeling algorithm is successful, the second step is to assign a probabilistic scam-likelihood score for the PAC of interest. To accomplish this, the labeled data can be used to train a supervised machine learning model (e.g. logistic regression). The output of the algorithm will be visible when interacting with the network graph.

Isolated Forest for anomaly detection was also thought to be appropriate to explore in this study as a one-step, label-and-score algorithm. Isolation Forest models are unsupervised and are able to quickly identify outliers in high dimensional data and assign anomaly scores. These can be used as "scam scores" for purposes of this study.

4.2 Approach

4.2.1 Data The expenditure data used in this project was downloaded from OpenSecrets. Although the data is generally clean, the expenditure data is not updated as frequently as the organization has "never had sufficient staff to properly work with the expenditure data" [15]. Therefore, for this study, the 2020 election cycle data was used as the most recent and complete data available. The dataset contains 131 million records of each expenditure made by a political committee during the 2020 campaign cycle and requires 27 GB of disk space. Once the data was extracted, it was joined on committees and candidates with additional data extracted from the FEC website (processed using polars library) using their API. The condensed transactions, committees, and candidates data was stored in an SQLite database. A schematic of the system architecture is provided in figure 1.

NetworkX, a python package, was utilized for building the graph. To best communicate relationships between all entities within the graph, we elected to produce a directed, non-bipartite graph.

A list of 31 known SPACs for 2020 was manually compiled for model verification through record-searching.

4.2.2 Modeling The models explored in this study were built using python's scikit-learn package. A k-means clustering algorithm (k=2) was chosen to label SPACs. Of the two resulting clusters, the smaller cluster was assumed to be SPACs due to the expected appearance frequency. The data was first explored to ensure that

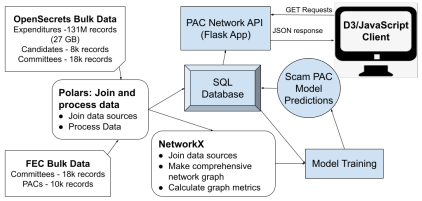

Figure 1: The system architecture

all assumptions are met for k-means clustering. This included normalizing features first, then ensuring feature independence and checking for outliers. Six k-means clustering models were constructed with model assumptions as guidelines:

- Base Model (BM): Includes all PAC expenditure categories (e.g. contributions, fundraising, refunds, etc.), a binary variable identifying if a PAC is older than 6 months (qualified) or younger than 6 months (nonqualified), and network graph metrics.
- k2: Excludes outliers.
- k3: Excludes binary variable.
- k4: Excludes some of the highly correlated expenditure categories and the binary variable.
- k5: Excludes all the highly correlated expenditure categories and the binary variable.
- k6: Same as k5 but excludes outliers.

All models were trained and evaluated on a 70:30 data split with 100 Monte Carlo simulations. Stratified sampling was used to maintain a balanced SPAC presence in the split and models were evaluated on the 30% test data split using a list of known SPACs in the data year. Due to the fact that k-means clustering algorithms are not generally not well-suited to handle binary or highly correlated variables, models k4 and k5 were expected to perform better because these variables were excluded. Surprisingly, these models underperformed and were unable to detect any known SPACs. Because SPAC detection was lower than anticipated across all models, the label outputs were unable to be used to train a supervised model. Model performance details can be found in section 5.

Isolated Forest models were also evaluated using the same setup as before with a 0.05 contamination parameter. The following models were constructed:

- i1: The same features as BM.

- i2: Includes a PAC designation (e.g. leadership PAC, unauthorized PAC, lobbyist PAC, etc.)
- i3: Excludes all highly correlated expenditure categories, PAC designation and binary features.
- i4: Excludes all centrality metrics, PAC designation and binary features.

For each Isolated Forest Monte Carlo simulation, the model produced an anomaly score for each observation. These scores were averaged for each PAC across all simulations. Although the intention was to use the anomaly score as a "scam score", the score range was thought to be potentially confusing to users of our website. Therefore, the anomaly scores were converted to binary classifications. To accomplish this, all scores across all simulations were averaged and three standard deviation units were subtracted to obtain a threshold. PACs with an anomaly score below this threshold were labeled SPACs.

4.2.3 Visualization A dynamic web application was created incorporating all aspects of our project (pulling data from the SQLite database, constructing a network, and responding to requests from the client for a specific PAC) using Flask—a Python based web application framework. Flask was chosen because it is lightweight, has built-in templates for rendering basic web pages, is simple to connect to SQLite databases, and is one of the most popular web frameworks both for industry and individual projects. Our app is hosted by Render at https://fec-app.onrender.com/.

Producing a section of network graph in a meaningful way to the viewer required a thoughtful approach as too many connections would be overwhelming, and too few would limit the information communicated. Therefore, an algorithm was developed for a user to be able to customize the visible network graph. Given a PAC P, expansion rate n, indirect transaction value t, and boolean selections for showing received payments and vendor expenditures, subgraph S is constructed using the following steps:

(1) Initialize S with node P
(2) Add the top n nodes receiving money from P to S, ordered by the amount received
(3) If showing vendor spending, add the top $2.5n$ entities receiving money from any node in the previous set of nodes, ordered by amount received
(4) If showing received payments, add the top n nodes that send money to S, ordered by amount sent
(5) Add the top t nodes that send money to nodes in S, ordered by the number of nodes receiving money, and secondarily by total amount sent
(6) Add all transactions between nodes in S as edges

The user can adjust algorithm parameters using the toggles at the top of the webpage. A screenshot of the toggles and resulting subgraph is provided in figure 2.

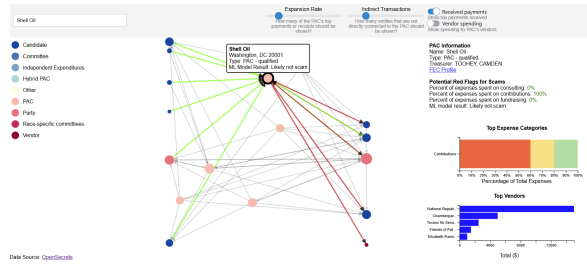

Figure 2: The completed website showing the graph and expenditure metrics for Shell Oil.

This study aimed to arm the user with the necessary information to assess whether a PAC can be trusted, so graphs and metrics associated with the node are provided when a user hovers over a PAC. These metrics were selected by "red flag" indicators reported by OpenSecrets [6]. Common indicators that a PAC may be a scam are large amounts of expenses being attributed to vague categories such as fundraising, administrative costs, or consulting. PAC expenses for the data year appear to the right of the graph and the user can easily determine if the expenses appear suspicious. To illustrate this, figure 3 shows the expense chart for the Police Officers Defense Alliance. This is a known SPAC, and hovering over the "fundraising" bar reveals that 63.3% of the total expenses were allocated towards fundraising fees, with the second-highest category of expenses being "Unclassifiable". These are red flag metrics and the user would be wise to exercise caution before contributing.

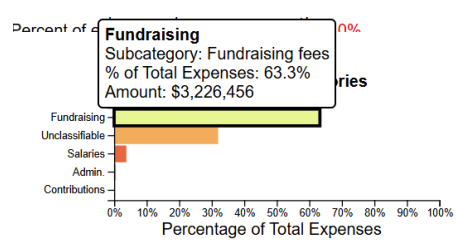

Figure 3: Expenses for Police Officers Defense Alliance, a known SPAC

We have also included a graph of the top vendors for each PAC so the user can easily see a summary of who is being paid by the PAC and corroborate any suspicions arising from the expense data. Typically genuine PACs include media companies or political candidates as top vendors. Conversely, SPACs will include private vendors that are either entirely unaffiliated with other PACs, very weakly connected in the network, or are connected with other known SPACs.

As a final help to users, we incorporated the binary classification output from our Isolated Forest model into the visualization by indicating whether a PAC is a likely scam or not. This indicator appears in the PAC label when a user interacts with a node along with the state and zip code (a common address between a PAC and vendor can also be concerning). In figure 4, the "ML Model Result" on the label for Police Officers Defense Alliance indicates this PAC may be a scam and the user should be careful. While model performance was sub-optimal, the output was still included to demonstrate the potential impact this information could have on user experience.

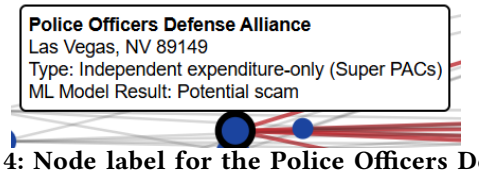

Figure 4: Node label for the Police Officers Defense Alliance

5 Experiments and Evaluation

Machine learning models were subject to traditional model evaluation methods intended to answer the following questions:

- Are the models a good fit given the data features?
- What features improve model performance, and which features hinder?
- How effective is the model at detecting SPACs?

Table 1: Mean results for all tested k-means clustering models. Descriptions of the models are in section 4.2.2. Silhouette scores for BM, k3, k4, k5 and k6 were 0.99, and the silhouette score for k2 was 1.0.

Model No.	Test Accuracy (%)	Inertia	DBI
BM	18.3 ∓10.4	7.2 ∓3.5	0.02 ∓0.0
k2	20.4 ∓10.4	0.37 ∓0.22	0.0 ∓0.0
k3	0.0 ∓0.0	3.1 ∓2.0	0.27 ∓0.31
k4	0.0 ∓0.0	1.9 ∓1.1	0.31 ∓0.28
k5	0.0 ∓0.0	1.4 ∓0.7	0.39 ∓0.25
k6	0.0 ∓0.0	0.04 ∓0.02	0.2 ∓0.17

Of all tested clustering models, the only models to detect any SPACs were the base model (BM) and k2 (see table 1). Both models utilized all available features; however, k2 (outliers removed) was able to detect 2.1% more SPACs on average. Both models demonstrated a strong, well-defined cluster with clear separation (see evaluation metrics in table 1). However, they include a binary variable that may introduce bias as k-means models typically struggle to handle binary variables effectively.

Table 2: Mean results for all tested Isolated Forest models. Descriptions of the models are in section 4.2.2. Kolmogorov–Smirnov test was also performed for each model with a test value of 1 and p<0.001 for all models.

No.	Test Accuracy (%)	Silhouette Score	DBI
i1	4.4 ∓6.1	0.80 ∓0.003	0.38 ∓0.02
i2	1.1 ∓3.3	0.44 ∓0.003	1.04 ∓0.03
i3	2.5 ∓4.5	0.98 ∓0.001	0.08 ∓0.01
i4	27.2 ∓13.5	0.98 ∓0.009	0.09 ∓0.02

Isolated Forest models were all able to detect some SPACs, with i4 detecting the highest number of SPACs as the overall best-performing model (see table 2). Its success was largely due to the exclusion of graph metrics from the features, resulting in a high silhouette score and relatively low DBI — indicators of a strong model fit. Across all tested models, removing highly correlated features reduced test accuracy.

Overall, i4 was able to identify on average 27% of the known SPACs in this study, and labeled additional PACs as scams (figure 5). While the models were strong according to goodness-of-fit metrics, the difficulty was

	Known Scam	
Labeled Scam	0	1
0	6186	22
1	262	9

Figure 5: Confusion matrix for Isolated Forest model 4

in evaluating SPAC detection due to the limited amount of known SPACs. Given this, it is possible that the model found SPACs that are undisclosed so far, thus making it more effective than test results suggest. The model also indicated a high rate of Type II errors, likely because some SPACs operate in a gray area, exhibiting partial scam characteristics, and were therefore undetected by the model.

We evaluated the effectiveness of our website with a user study designed to answer the following questions:

- Is the visualization easy for a user to interact with?
- Is the information provided in the visualization easy to interpret (i.e. an improvement over what already exists)?
- Can the user identify SPACs?

A group of 15 users were divided into two groups and tasked with using one of two methods to answer questions about the Police Officers Defense Alliance and Hallmark Cards PACs and determine whether they are scams. The first group (8 users) used existing websites (OpenSecrets and the FEC database), and the second group (7 users) used our PAC visualizer website to answer the questions.

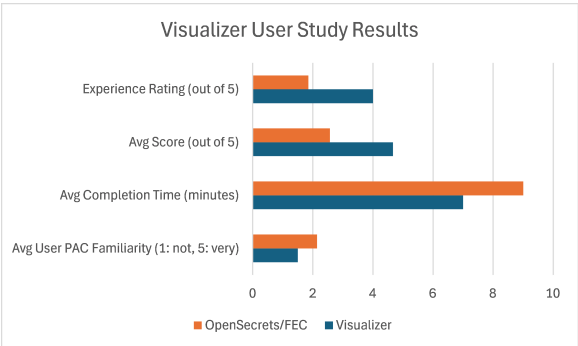

Figure 6: Results of the user study, where n=7 for the visualizer website, and n=8 for OpenSecrets/FEC website users

Although users of the existing websites self-reported a higher level of familiarity with PACs, these users on average took longer to answer the questions, answered fewer questions correctly, and generally were unable to consistently identify whether a PAC was a scam (figure 6). The users of our visualizer scored higher, completed the questions in a shorter time, and generally rated their experience as better. Further, these users were all able to correctly identify Hallmark Cards as a legitimate PAC and Police Officers Defense Alliance as a SPAC. The user studies show progress toward achieving our goal of creating an informative tool that is easy to navigate and an overall improvement over existing websites.

6 Discussion and Conclusion

In a field where nearly no published research exists, this study was able to explore SPAC detection with limited success. While the user studies showed that we were successful in alerting our website users to potential SPACs, we were unable to create an interpretable risk score using our model output. We also found that incorporating graph metrics in the models reduced model performance according to the list of known SPACs. However, given the limited number of disclosed SPACs, it is possible that the model's performance is understated here. Previously Li Zhao was able to use random forest models with greater detection success[13]. However, some of the PACs labeled as SPACs are found to be questionable by the authors of this study, raising questions about the true accuracy of the model.

The limits of explored machine learning models underscore the importance of the network graph. For example, although the model did not flag Heart Disease Network of America as a SPAC, exploring the graph reveals that this PAC shares common vendors with the Police Officers Defense Alliance SPAC. This connection is easily made in the graph, but it much more difficult to uncover sifting through data on the FEC website. In theory, identifying even one SPAC can reveal other suspicious activity through network connections.

Ultimately, we were able to meet the overarching objective of communicating FEC data to the average citizen in a way that is easy to comprehend.

This study can be expanded by experimenting with other machine learning models (or improving feature selection), including data from other election cycles, and refining what qualifies as a SPAC.

All team members have contributed a similar amount of effort to this project.

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